

Functional Requirement Document (FRD)

Farm-to-Customer Food Processing & Supply Chain Management System

Tagline

"Connecting Farmers, Processing Excellence, and Customers Through a Fully Traceable Digital Supply Chain."

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1. Project Overview

Module Overview

The **Farm-to-Customer Food Processing & Supply Chain Management System** is an integrated ERP platform designed to manage the complete agricultural product lifecycle from farmer onboarding to final customer delivery. The system digitizes procurement, production, quality assurance, warehouse operations, inventory management, sales, logistics, and customer

feedback while maintaining complete product traceability across every stage of the supply chain.

The platform enables multiple branches to operate independently under centralized supervision, ensuring standardized production processes, quality control, and operational transparency.

2. Project Objective

The primary objective of the system is to create a transparent, efficient, and fully traceable food processing ecosystem by digitally managing every business operation.

The system aims to:

- Digitize farmer registration and management.
 - Manage procurement of agricultural products.
 - Maintain farmer agreements and procurement records.
 - Monitor crop quality through field inspections.
 - Standardize production using predefined product recipes.
 - Maintain batch-level traceability.
 - Control warehouse inventory.
 - Manage customer orders and deliveries.
 - Improve quality assurance throughout manufacturing.
 - Generate business reports and analytics for decision-making.
-

3. Project Scope

The application covers the complete business process beginning with farmer onboarding and ending with customer delivery and feedback.

The platform manages:

- Farmer Registration
- Branch Operations

- Procurement Planning
- Agriculture Monitoring
- Production Management
- Recipe Management
- Quality Assurance
- Packaging
- Warehouse Operations
- Inventory Management
- Sales
- Dispatch
- Delivery Tracking
- Customer Feedback
- Business Analytics

The system supports multiple company branches while allowing the Super Admin to centrally monitor all branch activities through a unified dashboard.

4. Business Workflow

The application follows a structured operational workflow to ensure product traceability and process standardization throughout the food supply chain.

Stage 1 – Farm Sourcing & Planning

The process begins with farmer registration, agreement management, seed distribution, training, and regular field monitoring by agriculture experts.

Stage 2 – Procurement & Raw Material Control

After crop harvesting, raw materials are inspected, procured, assigned batch numbers, and stored in warehouses with complete traceability.

Stage 3 – Production & Quality Assurance

Approved raw materials undergo cleaning, grading, manufacturing, quality inspection, packaging, and finished goods storage using standardized product recipes.

Stage 4 – Sales & Distribution

Customer orders are processed after inventory validation, followed by billing, dispatch, shipment tracking, and successful delivery.

Stage 5 – Feedback & Continuous Improvement

Customers submit ratings, reviews, and complaints, while management analyzes business performance for continuous operational improvements.

5. User Roles

The application provides role-based access control to ensure every department manages only its authorized business functions.

5.1 Super Admin

Module Overview

The Super Admin manages the entire organization, including all branches, users, product recipes, inventory, production, reports, and business settings.

Permissions

- Dashboard Management
- Branch Management
- User Management
- Farmer Approval
- Recipe Management
- Procurement Monitoring
- Production Monitoring
- Warehouse Monitoring
- Inventory Control

- Sales Monitoring
 - Reports & Analytics
 - Master Settings
 - QR Traceability
 - Batch Tracking
 - Complaint Monitoring
-

5.2 Branch Manager

Module Overview

Each company branch is managed by a dedicated Branch Manager assigned by the Super Admin. The Branch Manager supervises all branch operations and ensures that manufacturing follows approved recipes and quality standards.

Permissions

- Branch Dashboard
 - Branch Staff Management
 - Procurement Monitoring
 - Production Monitoring
 - Warehouse Monitoring
 - Inventory Monitoring
 - Sales Monitoring
 - Branch Reports
 - Recipe Distribution
 - Quality Monitoring
-

5.3 Procurement Manager

Module Overview

The Procurement Manager is responsible for planning, inspecting, and purchasing agricultural raw materials from registered farmers.

Permissions

- Farmer List
 - Purchase Planning
 - Procurement Approval
 - Batch Assignment
 - Raw Material Inspection
 - Procurement Reports
-

5.4 Agriculture Expert

Module Overview

Agriculture Experts monitor crop growth and guide farmers throughout the cultivation process.

Permissions

- Farmer Visits
 - Crop Monitoring
 - Field Reports
 - Seed Distribution Records
 - Fertilizer Recommendations
 - Irrigation Advice
 - Training Records
-

5.5 Production Manager

Module Overview

The Production Manager controls all manufacturing activities and ensures production follows the approved recipes created by the Super Admin.

Permissions

- Production Planning
 - Recipe Execution
 - Batch Production
 - Machine Monitoring
 - Process Monitoring
 - Production Reports
-

5.6 QA Manager

Module Overview

The QA Manager ensures that all raw materials and finished products meet defined quality standards before market release.

Permissions

- Quality Inspection
 - Batch Approval
 - Product Release
 - Reject Batch
 - Rework Approval
 - QA Reports
-

5.7 Warehouse Manager

Module Overview

The Warehouse Manager controls storage, stock movement, and inventory availability across warehouses.

Permissions

- Stock Entry
- Batch Storage
- Inventory Reports

- Warehouse Transfers
 - Dispatch Preparation
 - Stock Validation
-

5.8 Sales Team

Module Overview

The Sales Team manages customer orders, invoices, payments, and sales records.

Permissions

- Customer Management
 - Order Management
 - Invoice Generation
 - Payment Tracking
 - Sales Reports
 - Customer Communication
-

5.9 Logistics Team

Module Overview

The Logistics Team is responsible for dispatching products and ensuring timely customer deliveries.

Permissions

- Shipment Assignment
 - Route Management
 - Delivery Tracking
 - Delivery Confirmation
 - Proof of Delivery
-

5.10 Customer

Module Overview

Customers can purchase products, track deliveries, verify product authenticity, and provide valuable feedback.

Permissions

- Browse Products
 - Place Orders
 - Track Orders
 - Scan QR Code
 - View Product Traceability
 - Submit Ratings
 - Raise Complaints
 - View Order History
-

6. Branch Management Module

Module Overview

The Branch Management Module enables the Super Admin to create and manage multiple company branches while maintaining centralized control over branch operations.

6.1 Branch Creation

The Super Admin can create multiple company branches by providing branch name, location, address, contact information, and operational details.

6.2 Branch Manager Assignment

The Super Admin can assign a dedicated Branch Manager to manage the day-to-day operations of each branch.

6.3 Branch Dashboard

Each Branch Manager has access to a dashboard displaying procurement activities, production status, warehouse inventory, sales performance, and pending tasks specific to their branch.

6.4 Branch Performance

The system tracks branch-wise production, sales, inventory utilization, procurement volume, and operational efficiency.

6.5 Branch Reports

Branch Managers can generate reports for procurement, production, warehouse, inventory, and sales activities, while the Super Admin can view consolidated reports across all branches.

7. Farmer Management Module

Module Overview

The Farmer Management Module is the foundation of the system, allowing the organization to register, verify, and manage farmers associated with different branches. It maintains complete farmer records, procurement history, crop information, and payment details for better planning and traceability.

7.1 Farmer Registration

The system allows Procurement Managers or authorized branch staff to register new farmers by capturing complete personal, farming, and banking information.

Registration Details

Personal Information

- Farmer Name
- Mobile Number
- Aadhaar Number
- PAN Number

- Family Details

Address Information

- Village
- District
- State
- Complete Address
- GPS Location

Farm Information

- Farm Size
- Land Type
- Irrigation Type
- Crop Details

Bank Details

- Account Holder Name
- Bank Name
- Account Number
- IFSC Code

7.2 Farmer Verification

Every newly registered farmer must be verified before participating in procurement activities.

Features

- Approve Farmer
- Reject Farmer
- Request Additional Documents
- Verification Remarks
- Verification History

7.3 Farmer Profile

The system maintains a centralized farmer profile containing all historical and operational information.

Profile Information

- Personal Details
 - Farm Information
 - Crop History
 - Procurement History
 - Payment History
 - Training History
 - Field Visit History
 - Quality Rating
 - Active Agreements
-

7.4 Farmer Search

Authorized users can quickly locate farmers using multiple search filters.

Search Filters

- Farmer Name
- Village
- District
- State
- Crop
- Branch
- Active Status
- Quality Rating

7.5 Farmer Status Management

Administrators can control the operational status of farmers.

Status

- Active
- Inactive
- Blacklisted
- Suspended

7.6 Farmer Performance

The system automatically tracks farmer performance based on procurement quality and delivery consistency.

Performance Parameters

- Crop Quality
- Delivery Timeliness
- Procurement Quantity
- Complaint Records
- Overall Rating

8. Farmer ID & Traceability Module

Module Overview

The Farmer ID & Traceability Module provides every registered farmer with a unique identity to ensure complete traceability from cultivation to the finished product.

8.1 Unique Farmer ID

Every approved farmer receives an automatically generated unique identification number.

Example

8.2 QR Code Generation

The system generates a QR Code linked to the farmer profile for quick identification and record retrieval.

8.3 Barcode Generation

A barcode is also generated for warehouse operations and procurement activities.

8.4 Traceability Information

The farmer ID is linked with:

- Farm Location
 - Crop Details
 - Procurement Records
 - Production Batch
 - Product Batch
 - Customer QR Code
-

9. Agreement Management Module

Module Overview

The Agreement Management Module maintains procurement agreements between the company and registered farmers before the cultivation season begins.

9.1 Agreement Creation

Procurement Managers can create agreements specifying procurement terms and expected crop supply.

Agreement Details

- Farmer Name
 - Crop Name
 - Variety
 - Expected Quantity
 - Purchase Rate
 - Agreement Date
 - Harvest Date
 - Quality Standards
-

9.2 Agreement Status

The system tracks agreement progress throughout the farming cycle.

Status

- Pending
 - Active
 - Completed
 - Cancelled
-

9.3 Agreement History

All previous agreements remain available for future reference and audit purposes.

10. Seed & Input Distribution Module

Module Overview

The Seed & Input Distribution Module records the distribution of seeds and agricultural inputs provided by the company to farmers.

10.1 Seed Distribution

Authorized Agriculture Experts can distribute certified seeds to registered farmers.

Distribution Details

- Seed Name
 - Seed Variety
 - Quantity
 - Distribution Date
 - Farmer Name
 - Batch Number
-

10.2 Distribution Tracking

The system records every distribution transaction to maintain inventory and farmer-wise usage history.

10.3 Distribution History

Users can review complete seed distribution records for each farmer.

11. Training Management Module

Module Overview

The Training Management Module enables the organization to conduct educational programs for farmers and maintain complete training records.

11.1 Training Schedule

Agriculture Experts can create training sessions based on seasonal farming practices and crop requirements.

11.2 Attendance Management

Attendance is recorded for every farmer participating in the training program.

11.3 Training Materials

Training documents can be uploaded in multiple formats.

Supported Files

- PDF Documents
 - Images
 - Presentations
 - Videos
-

11.4 Training History

The application stores all completed training sessions attended by each farmer.

12. Field Monitoring Module

Module Overview

The Field Monitoring Module enables Agriculture Experts to regularly inspect farms, monitor crop conditions, and provide technical recommendations to improve agricultural productivity.

12.1 Field Visit

Agriculture Experts can schedule and record farm visits throughout the crop cultivation period.

Visit Information

- Visit Date
 - Farmer Name
 - Crop Name
 - Branch
 - Expert Name
-

12.2 Crop Monitoring

Experts can record the current crop condition during each field visit.

Crop Details

- Crop Growth Stage
 - Crop Health
 - Pest Status
 - Disease Observation
-

12.3 Agriculture Recommendations

Experts can provide recommendations for improving crop quality.

Recommendations

- Fertilizer Advice
 - Irrigation Advice
 - Pest Control Suggestions
 - Harvest Preparation
-

12.4 Yield Prediction

Agriculture Experts can estimate expected production quantity based on crop growth and field observations.

12.5 Document Upload

Field inspection records can include supporting documents.

Supported Uploads

- Crop Photos
- Field Images
- PDF Reports

- Inspection Documents
-

12.6 Field Visit History

The system maintains a complete history of all field inspections for every registered farmer.

12.7 Field Report

A detailed field report is automatically generated after every inspection, helping procurement and production teams prepare for upcoming harvests.

13. Procurement Management Module

Module Overview

The Procurement Management Module manages the complete process of purchasing agricultural produce from registered farmers. It ensures that only approved and quality-compliant raw materials enter the production process through a structured inspection and approval workflow.

13.1 Procurement Planning

The Procurement Manager can prepare procurement plans based on production requirements, seasonal demand, and farmer agreements.

Features

- Crop-wise Procurement Planning
 - Branch-wise Procurement Planning
 - Quantity Planning
 - Procurement Schedule
 - Procurement Status
-

13.2 Harvest Inspection

Before collecting raw materials, every harvest is inspected to ensure it meets the company's quality standards.

Inspection Checklist

- Moisture Level
 - Foreign Matter
 - Grain Size
 - Grain Color
 - Smell
 - Physical Damage
-

13.3 Quality Inspection Documents

Inspectors can upload supporting documents during procurement inspection.

Supported Files

- Crop Images
 - Inspection Photos
 - PDF Reports
 - Quality Certificates
-

13.4 Inspection Result

After inspection, the Procurement Manager records the final quality decision.

Status

- Approved
 - Rejected
 - Hold for Re-Inspection
-

13.5 Procurement Approval

Only approved harvests can proceed to the raw material collection stage for manufacturing.

14. Raw Material Collection Module

Module Overview

The Raw Material Collection Module records the collection of approved agricultural products from farmers and maintains complete purchase information for traceability.

14.1 Collection Entry

The system records every procurement transaction completed with the farmer.

Collection Details

- Farmer Name
 - Crop Name
 - Collection Date
 - Branch
 - Collection Location
-

14.2 Weight Measurement

The exact quantity of collected raw material is recorded during procurement.

Weight Information

- Gross Weight
 - Net Weight
 - Unit of Measurement
-

14.3 Purchase Details

The purchase value is calculated based on the agreed procurement rate.

Purchase Information

- Purchase Price
 - Quantity
 - Total Amount
 - Payment Status
-

14.4 Receipt Generation

The system generates a procurement receipt for both the farmer and the company.

Receipt Includes

- Receipt Number
 - Farmer Details
 - Product Details
 - Quantity
 - Purchase Amount
 - Collection Date
-

14.5 Collection History

Every procurement transaction is stored for audit, reporting, and future reference.

15. Batch Management Module

Module Overview

The Batch Management Module generates unique batch numbers for every collected raw material lot, ensuring complete traceability throughout manufacturing and delivery.

15.1 Raw Material Batch Generation

The system automatically generates a unique batch number for every approved procurement.

Example

15.2 Batch Information

Each batch stores complete procurement and traceability information.

Batch Details

- Batch Number
 - Farmer Name
 - Crop Name
 - Collection Date
 - Quantity
 - Warehouse
 - Branch
-

15.3 Batch Tracking

The application tracks every batch throughout the manufacturing lifecycle.

Tracking Stages

- Raw Material Collection
 - Warehouse Storage
 - Production
 - Packaging
 - Finished Goods
 - Customer Delivery
-

15.4 Batch Status

The system maintains the current operational status of every batch.

Status

- Collected
 - Stored
 - Under Production
 - Packaged
 - Dispatched
 - Delivered
-

15.5 QR & Barcode Association

Each batch is linked with QR Codes and Barcodes for faster identification and product traceability.

16. Raw Material Warehouse Module

Module Overview

The Raw Material Warehouse Module manages the storage and movement of all procured agricultural products before production begins.

16.1 Stock Entry

Approved raw material batches are entered into the warehouse inventory upon arrival.

Stock Information

- Batch Number
 - Product Name
 - Quantity
 - Storage Date
 - Warehouse Location
-

16.2 Batch Storage

The system stores inventory batch-wise to maintain complete traceability and stock accuracy.

16.3 Warehouse Allocation

Warehouse Managers can assign storage locations to different raw material batches based on available capacity and product type.

16.4 Stock Transfer

Raw materials can be transferred between warehouses or storage sections while maintaining complete transfer records.

16.5 Re-Stock Management

The system supports restocking activities whenever inventory is replenished or transferred back into storage.

16.6 Warehouse Status

Managers can monitor real-time warehouse occupancy and available storage capacity.

16.7 Warehouse Reports

The system generates warehouse reports showing:

- Current Stock
 - Batch-wise Inventory
 - Stock Movement
 - Incoming Stock
 - Outgoing Stock
-

17. Inventory Management Module

Module Overview

The Inventory Management Module provides real-time visibility of raw material stock available across all warehouses and branches. It helps prevent stock shortages, overstocking, and inventory discrepancies.

17.1 Available Stock

The system displays the total quantity of raw materials currently available for production.

17.2 Reserved Stock

Raw materials allocated for upcoming production batches are marked as reserved to avoid accidental usage.

17.3 Stock Movement

All stock movements are automatically recorded.

Movement Types

- Stock In
 - Stock Out
 - Stock Transfer
 - Stock Adjustment
-

17.4 Low Stock Alert

The application automatically notifies warehouse managers when inventory reaches the minimum stock level.

17.5 Inventory History

Every inventory transaction is stored with timestamps and user details for auditing purposes.

17.6 Branch-wise Inventory

The Super Admin can monitor inventory levels across all company branches from a centralized dashboard.

18. Cleaning & Grading Module

Module Overview

Before raw materials enter the production process, they undergo cleaning and grading to remove impurities and ensure only quality-approved materials are used for manufacturing.

18.1 Cleaning Process

The system records the cleaning process performed on each raw material batch.

Activities

- Dust Removal
 - Stone Removal
 - Foreign Material Removal
 - Impurity Separation
-

18.2 Grading Process

Raw materials are graded according to predefined quality standards.

Grading Parameters

- Grain Size
 - Color
 - Texture
 - Moisture Level
 - Purity
-

18.3 Quality Verification

The QA team verifies that cleaned and graded raw materials meet production standards before approving them for manufacturing.

18.4 Cleaning Records

The application maintains detailed records of every cleaning and grading activity, including the responsible operator, date, batch number, and inspection results.

19. Recipe (Formula) Management Module

Module Overview

The Recipe Management Module standardizes the manufacturing process by defining the exact ingredients, proportions, and production steps required for each product. Recipes are created only by the Super Admin and are shared with branches to ensure consistent product quality across all manufacturing units.

19.1 Recipe Creation

The Super Admin can create a recipe for every product manufactured by the company.

Recipe Details

- **Product Name**
 - **Product Category**
 - **Recipe Code**
 - **Version Number**
 - **Product Description**
-

19.2 Ingredient Management

Each recipe contains the complete list of raw materials required for manufacturing.

Details

- **Raw Material Name**

- Quantity
 - Unit of Measurement
 - Ingredient Percentage
 - Batch Requirement
-

19.3 Production Formula

The system stores the exact manufacturing formula required for product preparation.

Formula Information

- Mixing Ratio
 - Processing Sequence
 - Grinding Instructions
 - Roasting Instructions
 - Oil Extraction Process (If Applicable)
 - Packaging Instructions
-

19.4 Recipe Approval

Only approved recipes can be used during production.

Status

- Draft
 - Pending Approval
 - Approved
 - Inactive
-

19.5 Branch Recipe Distribution

Once approved, the recipe is automatically available to the assigned branch.

The Branch Manager shares the approved recipe with the Production Manager before manufacturing begins, ensuring all branches follow the same production standards.

19.6 Recipe Revision History

The system maintains version history whenever recipes are modified to ensure complete traceability of production changes.

20. Production Management Module

Module Overview

The Production Management Module controls the complete manufacturing lifecycle, from production planning to finished product generation. Every production batch is linked with raw material batches to maintain complete traceability.

20.1 Production Planning

Production Managers can schedule manufacturing based on customer demand, inventory availability, and production capacity.

Planning Details

- **Product Name**
 - **Production Date**
 - **Production Quantity**
 - **Assigned Branch**
 - **Assigned Production Line**
-

20.2 Production Batch Creation

A unique production batch is created before manufacturing begins.

Batch Details

- **Production Batch Number**

- **Product Name**
 - **Recipe Used**
 - **Production Date**
 - **Production Status**
-

20.3 Single Grain Production

The system supports manufacturing products made from a single raw material.

Examples

- **Wheat Flour**
- **Rice Flour**
- **Millet Flour**

Processes

- **Cleaning**
 - **Grinding**
 - **Roasting**
 - **Oil Extraction**
 - **Packaging**
-

20.4 Multi Grain Production

For blended products, the system follows the approved production recipe.

Production Activities

- **Ingredient Selection**
- **Weight Measurement**
- **Mixing**
- **Grinding**
- **Blending**

- Homogeneity Check
 - Packing
-

20.5 Raw Material Consumption

The application records the quantity of raw materials consumed during each production batch.

Details

- Raw Material Batch
 - Quantity Used
 - Remaining Stock
 - Production Loss
-

20.6 Machine Allocation

Production Managers assign machines required for manufacturing each production batch.

Information

- Machine Name
 - Machine Number
 - Operator Name
 - Production Line
-

20.7 Production Monitoring

The system monitors every production stage until completion.

Status

- Planned
- In Progress
- Completed

- Paused
 - Cancelled
-

20.8 Production Records

All manufacturing activities are automatically stored for future reference and auditing.

21. Quality Management Module

Module Overview

The Quality Management Module ensures that quality standards are maintained throughout procurement, manufacturing, and packaging. Quality inspections are conducted at every stage before products move to the next process.

21.1 Raw Material Quality Inspection

The QA team verifies the quality of raw materials before production begins.

Inspection Parameters

- Moisture
 - Purity
 - Grain Size
 - Color
 - Foreign Matter
 - Odor
-

21.2 In-Process Quality Inspection

Quality checks are conducted during manufacturing to ensure production follows approved standards.

Inspection Parameters

- Ingredient Ratio

- **Mixing Accuracy**
 - **Grinding Quality**
 - **Temperature**
 - **Product Consistency**
-

21.3 Finished Goods Inspection

Finished products are inspected before packaging and warehouse storage.

Inspection Parameters

- **Product Appearance**
 - **Product Weight**
 - **Packaging Quality**
 - **Label Accuracy**
 - **Shelf Life Verification**
-

21.4 Quality Decision

The QA Manager records the inspection result for every batch.

Status

- **Pass**
 - **Fail**
 - **Rework Required**
-

21.5 Batch Approval

Only approved batches are released for packaging and warehouse storage.

Rejected batches cannot proceed until corrective actions are completed.

21.6 QA Reports

The system generates detailed quality reports for every inspection conducted during the production lifecycle.

22. Packaging Management Module

Module Overview

The Packaging Management Module prepares finished products for storage and distribution by generating packaging information, product labels, QR codes, and batch identification.

22.1 Finished Product Batch

The system generates a unique finished goods batch number for every packaged product.

22.2 Product Label Generation

The application automatically creates product labels containing all mandatory product information.

Label Information

- **Product Name**
 - **Batch Number**
 - **Manufacturing Date**
 - **Expiry Date**
 - **Net Weight**
 - **MRP**
 - **Packaging Date**
 - **Shelf Life**
-

22.3 QR Code Generation

A QR Code is generated for every packaged product to provide complete product traceability.

After scanning the QR Code, customers can view:

- Farmer Details
 - Farm Location
 - Procurement Batch
 - Production Batch
 - Packaging Date
 - Manufacturing Date
-

22.4 Barcode Generation

Each packaged product also receives a barcode for warehouse operations, inventory management, and retail billing.

22.5 Packaging Types

The system supports different packaging formats based on product requirements.

Examples

- Pouch
 - Box
 - Bottle
 - Jar
 - Sack
-

22.6 Packaging Records

The application stores complete packaging history for every finished product batch.

23. Finished Goods Warehouse Module

Module Overview

The Finished Goods Warehouse Module manages packaged products after manufacturing until they are dispatched for customer orders.

23.1 Stock Entry

Finished products are automatically added to warehouse inventory after successful packaging.

23.2 Available Stock

Displays the quantity of products available for customer orders.

23.3 Reserved Stock

Products allocated to confirmed orders are marked as reserved until dispatch.

23.4 Warehouse Transfer

Finished goods can be transferred between branches or warehouses while maintaining complete transfer records.

23.5 Batch-wise Storage

Products are stored batch-wise to maintain proper inventory management and complete traceability.

23.6 Warehouse Reports

The system generates warehouse reports including:

- **Available Stock**
 - **Reserved Stock**
 - **Batch-wise Inventory**
 - **Transfer History**
 - **Dispatch Ready Stock**
-

24. Sales Management Module

Module Overview

The Sales Management Module manages the complete sales lifecycle, from customer order placement to invoice generation and order fulfillment. It provides real-time visibility into sales performance and customer transactions across all branches.

24.1 Customer Management

The system maintains a centralized database of all customers to streamline order processing and communication.

Customer Details

- **Customer Name**
 - **Mobile Number**
 - **Email Address**
 - **Billing Address**
 - **Shipping Address**
 - **Order History**
 - **Payment History**
-

24.2 Order Management

Sales executives can create and manage customer orders received through the User Application.

Order Details

- **Order Number**
- **Customer Name**
- **Product Details**
- **Quantity**
- **Order Date**
- **Delivery Address**

- **Payment Status**
 - **Order Status**
-

24.3 Sales Dashboard

The Sales Team can monitor daily sales activities through a centralized dashboard.

Dashboard Information

- **Total Orders**
 - **Pending Orders**
 - **Completed Orders**
 - **Cancelled Orders**
 - **Total Revenue**
 - **Branch-wise Sales**
-

24.4 Sales Reports

The system automatically generates reports for business analysis.

Available Reports

- **Daily Sales Report**
 - **Monthly Sales Report**
 - **Product-wise Sales**
 - **Branch-wise Sales**
 - **Customer-wise Sales**
 - **Revenue Report**
-

25. Inventory Validation Module

Module Overview

Before confirming any customer order, the system validates inventory availability to prevent over-selling and ensure accurate order fulfillment.

25.1 Stock Availability Check

The application verifies whether sufficient stock is available before accepting an order.

25.2 Reserved Stock Validation

The system checks whether the requested inventory has already been reserved for another customer.

25.3 Warehouse Validation

If multiple warehouses exist, the system identifies the nearest warehouse with available stock for faster dispatch.

25.4 Order Approval

Orders proceed for billing only after successful inventory validation.

Status

- **Stock Available**
 - **Partially Available**
 - **Out of Stock**
-

26. Billing & Invoice Module

Module Overview

The Billing Module generates invoices, records payments, and maintains financial records for every completed transaction.

26.1 Invoice Generation

Invoices are automatically generated after successful order confirmation.

Invoice Includes

- **Invoice Number**
 - **Customer Details**
 - **Product Details**
 - **Quantity**
 - **Unit Price**
 - **GST Details**
 - **Total Amount**
 - **Payment Method**
-

26.2 Payment Management

The system records customer payments against invoices.

Payment Modes

- **Cash**
 - **UPI**
 - **Debit Card**
 - **Credit Card**
 - **Net Banking**
-

26.3 Receipt Generation

After successful payment, a digital receipt is generated and shared with the customer.

26.4 Payment Status

The application tracks payment progress for every order.

Status

- Pending
 - Paid
 - Failed
 - Refunded
-

27. Dispatch Management Module

Module Overview

The Dispatch Management Module manages shipment preparation and assigns delivery partners for customer orders.

27.1 Dispatch Planning

Warehouse Managers prepare confirmed orders for shipment.

Details

- Order Number
 - Dispatch Date
 - Warehouse
 - Assigned Staff
-

27.2 Delivery Partner Assignment

The Logistics Team assigns the most suitable delivery partner for each shipment.

Assignment Details

- Delivery Partner
 - Vehicle Information
 - Driver Name
 - Contact Number
-

27.3 Shipment Status

Every shipment is monitored until delivery.

Status

- Ready for Dispatch
 - Dispatched
 - In Transit
 - Out for Delivery
 - Delivered
-

27.4 Dispatch History

Complete dispatch records are maintained for future reference and reporting.

28. Delivery Tracking Module

Module Overview

The Delivery Tracking Module allows customers and internal teams to monitor shipment progress in real time until successful delivery.

28.1 Live Delivery Status

Customers can view the latest delivery status directly from the application.

28.2 Estimated Delivery Date

The system displays the expected delivery date based on shipment information.

28.3 OTP Verification

To ensure secure delivery, customers receive an OTP that must be verified before handing over the order.

28.4 Proof of Delivery

The delivery partner uploads proof after successful delivery.

Supported Proof

- Customer Signature
 - Delivery Photo
 - OTP Verification
 - Delivery Timestamp
-

28.5 Delivery History

Customers can access their complete delivery history through their account.

29. Customer Module

Module Overview

The Customer Module provides a simple and transparent platform for purchasing products, tracking deliveries, and accessing product traceability information.

29.1 Product Browsing

Customers can browse available products with detailed descriptions, pricing, nutritional information, and packaging details.

29.2 Place Order

Customers can add products to their cart and place orders through the mobile or web application.

29.3 Order History

The application stores all previous customer orders for easy reference.

29.4 Track Orders

Customers can monitor the current status of their orders until delivery.

29.5 Product Details

Each product displays:

- **Product Name**
 - **Product Images**
 - **Product Description**
 - **Ingredients**
 - **Net Weight**
 - **MRP**
 - **Manufacturing Date**
 - **Expiry Date**
-

30. Product Traceability Module

Module Overview

The Product Traceability Module enables complete transparency by allowing customers and administrators to trace every product back to its source using a QR Code.

30.1 QR Code Scanning

Customers can scan the QR Code printed on the product packaging to access detailed product information.

30.2 Farmer Information

The system displays the farmer associated with the raw materials used in the product.

Information Includes

- **Farmer ID**

- Farmer Name
 - Village
 - District
-

30.3 Procurement Details

Customers can view procurement-related information.

Details

- Procurement Date
 - Raw Material Batch Number
 - Collection Branch
-

30.4 Production Details

Displays complete manufacturing information.

Information

- Production Batch Number
 - Manufacturing Date
 - Recipe Version
 - Production Branch
-

30.5 Quality Information

Customers can verify that the product has successfully passed quality inspections.

Displays

- QA Approval Status
 - Inspection Date
 - Quality Certification
-

30.6 Packaging Information

The system displays packaging-related information.

Includes

- Packaging Date
 - Expiry Date
 - Shelf Life
 - Packaging Type
-

30.7 Delivery Information

The QR traceability record also displays dispatch information for complete supply chain transparency.

31. Feedback Management Module

Module Overview

The Feedback Management Module allows customers to share their experience after receiving the product. The collected feedback helps improve product quality, delivery services, and overall customer satisfaction.

31.1 Product Rating

Customers can rate purchased products based on their experience.

Rating Criteria

- Product Quality
 - Freshness
 - Packaging
 - Taste
 - Overall Satisfaction
-

31.2 Product Review

Customers can write detailed reviews explaining their experience with the product.

Features

- **Written Review**
 - **Product Images (Optional)**
 - **Purchase Verification**
 - **Review Date**
-

31.3 Delivery Rating

Customers can rate the delivery experience separately.

Parameters

- **Delivery Time**
 - **Delivery Executive**
 - **Packaging Condition**
 - **Overall Delivery Experience**
-

31.4 Feedback History

The application maintains a complete history of all customer ratings and reviews for future reference.

32. Customer Support & Complaint Management

Module Overview

The Customer Support Module enables customers to raise complaints regarding products or deliveries. Every complaint follows a structured workflow until resolution.

32.1 Complaint Registration

Customers can register complaints directly from the application.

Complaint Types

- **Damaged Product**
 - **Wrong Product**
 - **Packaging Issue**
 - **Quality Issue**
 - **Late Delivery**
 - **Other**
-

32.2 Complaint Details

Customers provide detailed information while submitting a complaint.

Details

- **Order Number**
 - **Product Name**
 - **Complaint Description**
 - **Supporting Images**
 - **Supporting Documents**
-

32.3 Complaint Assignment

The system automatically assigns complaints to the appropriate support executive for investigation.

32.4 Complaint Workflow

Every complaint passes through predefined stages.

Status

- **New**
- **Assigned**

- Under Investigation
 - In Progress
 - Resolved
 - Closed
-

32.5 Resolution Management

Support executives record investigation findings and update customers regarding the resolution.

32.6 Complaint History

All complaints remain available for audit purposes and customer reference.

33. Notification Module

Module Overview

The Notification Module keeps all users informed about important business activities through real-time alerts and automated notifications.

33.1 Farmer Notifications

Farmers receive notifications regarding:

- Registration Approval
 - Agreement Status
 - Training Schedule
 - Seed Distribution
 - Procurement Schedule
 - Payment Updates
-

33.2 Branch Notifications

Branch Managers receive alerts related to:

- **Procurement Updates**
 - **Production Status**
 - **Low Inventory**
 - **Quality Approval**
 - **Warehouse Transfers**
-

33.3 Production Notifications

Production Managers receive notifications for:

- **Production Schedule**
 - **Recipe Updates**
 - **Batch Completion**
 - **QA Approval**
 - **Machine Alerts**
-

33.4 Warehouse Notifications

Warehouse Managers receive alerts regarding:

- **Low Stock**
 - **New Stock Arrival**
 - **Warehouse Transfer**
 - **Dispatch Preparation**
-

33.5 Customer Notifications

Customers receive notifications for:

- **Order Confirmation**
- **Invoice Generated**

- **Dispatch Status**
 - **Delivery Updates**
 - **Complaint Status**
-

Notification Channels

- **In-App Notification**
 - **SMS Notification**
 - **Email Notification**
-

34. Reports & Analytics Module

Module Overview

The Reports & Analytics Module provides real-time business insights to management through interactive dashboards and downloadable reports.

Farmer Reports

- **Registered Farmers**
 - **Active Farmers**
 - **Crop-wise Farmers**
 - **Village-wise Farmers**
 - **Farmer Performance Report**
-

Procurement Reports

- **Daily Procurement**
- **Crop-wise Procurement**
- **Farmer-wise Procurement**
- **Purchase Cost Report**

Production Reports

- **Daily Production**
- **Product-wise Production**
- **Recipe Usage**
- **Production Efficiency**
- **Machine Utilization**

Quality Reports

- **QA Pass Report**
- **QA Fail Report**
- **Rework Report**
- **Batch Inspection Report**

Warehouse Reports

- **Current Stock**
- **Available Inventory**
- **Reserved Inventory**
- **Warehouse Transfers**

Sales Reports

- **Daily Sales**
- **Monthly Sales**
- **Product-wise Sales**
- **Customer-wise Sales**
- **Branch-wise Sales**

Delivery Reports

- **Pending Deliveries**
- **Completed Deliveries**
- **Delivery Performance**
- **Delayed Shipments**

Financial Reports

- **Revenue Report**
- **Procurement Cost**
- **Production Cost**
- **Profit Analysis**

Branch Reports

The Super Admin can compare branch performance based on:

- **Production**
- **Procurement**
- **Sales**
- **Inventory**
- **Revenue**

35. Upload & Document Management Module

Module Overview

This module provides a centralized document repository for uploading and managing all business-related files across different departments.

Supported Upload Formats

- **PDF**
 - **Images**
 - **Excel Files**
 - **Word Documents**
-

Documents Managed

- **Farmer Documents**
 - **Aadhaar & PAN**
 - **Agreements**
 - **Training Materials**
 - **Field Inspection Reports**
 - **Quality Certificates**
 - **Procurement Reports**
 - **Production Documents**
 - **Packaging Certificates**
-

Features

- **Upload**
 - **Preview**
 - **Download**
 - **Replace**
 - **Delete**
 - **Version History**
-